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Philosophers of Science Facilitated Intellectual Exchange Between Developmental Approaches in Ethology and Evolutionary Biology

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ABSTRACT

Developmental processes have often been granted a secondary role in ethology and evolutionary biology. Yet, in ethology, developmental psychobiology was established as an alternative to the instinct concept, emphasizing the importance of environmental influences for the development of species-typical behavior. Similarly, in evolutionary biology, the extended evolutionary synthesis rejects gene-centric perspectives to recognize a crucial role of development for adaptation and diversification. In this paper, we study the connections between these developmentalist perspectives. We focus specifically on West and King's ontogenetic niche concept and its relationship to the constructive development principle of the extended evolutionary synthesis. Using tools from computational history and philosophy of science, we provide evidence that the philosophical tradition of developmental systems theory mediated the transfer of the ontogenetic niche concept into evolutionary biology. We conclude with reflections about the role of philosophers of science as bridge-builders between research fields and perspectives.

1 | Introduction

The courtship of the satin bowerbird (*Ptilonorhynchus violaceus*) is remarkable. Males construct an avenue bower on the forest floor from carefully arranged sticks, forming a narrow corridor, which they decorate with blue feathers, flowers, and other objects (Figure 1A; see Borgia 1985). When a female visits, the male performs a highly stylized display of movements, vocalizations, and body vibrations. Mating may follow, depending on the female's evaluation of the esthetic structure and the male's behavioral skill.

These striking behaviors raise fundamental questions. How do males acquire the skills required to build and display? What criteria guide female choice? How did these elaborate male–female interactions evolve, and what role did behavior and aesthetic preferences play in this process? More generally, how do traits arise over the course of an organism's life, and how do such processes influence evolutionary trajectories?

These questions draw attention to development. Yet, this process has long occupied a secondary explanatory role in both ethology and evolutionary biology. Ethologists historically downplayed ontogenetic trajectories by treating species-typical behaviors as largely innate (Blumberg 2017; Lorenz and Tinbergen 1938; Tinbergen 1951), while evolutionary biologists often regarded development as the mere unfolding of a genetic program, and

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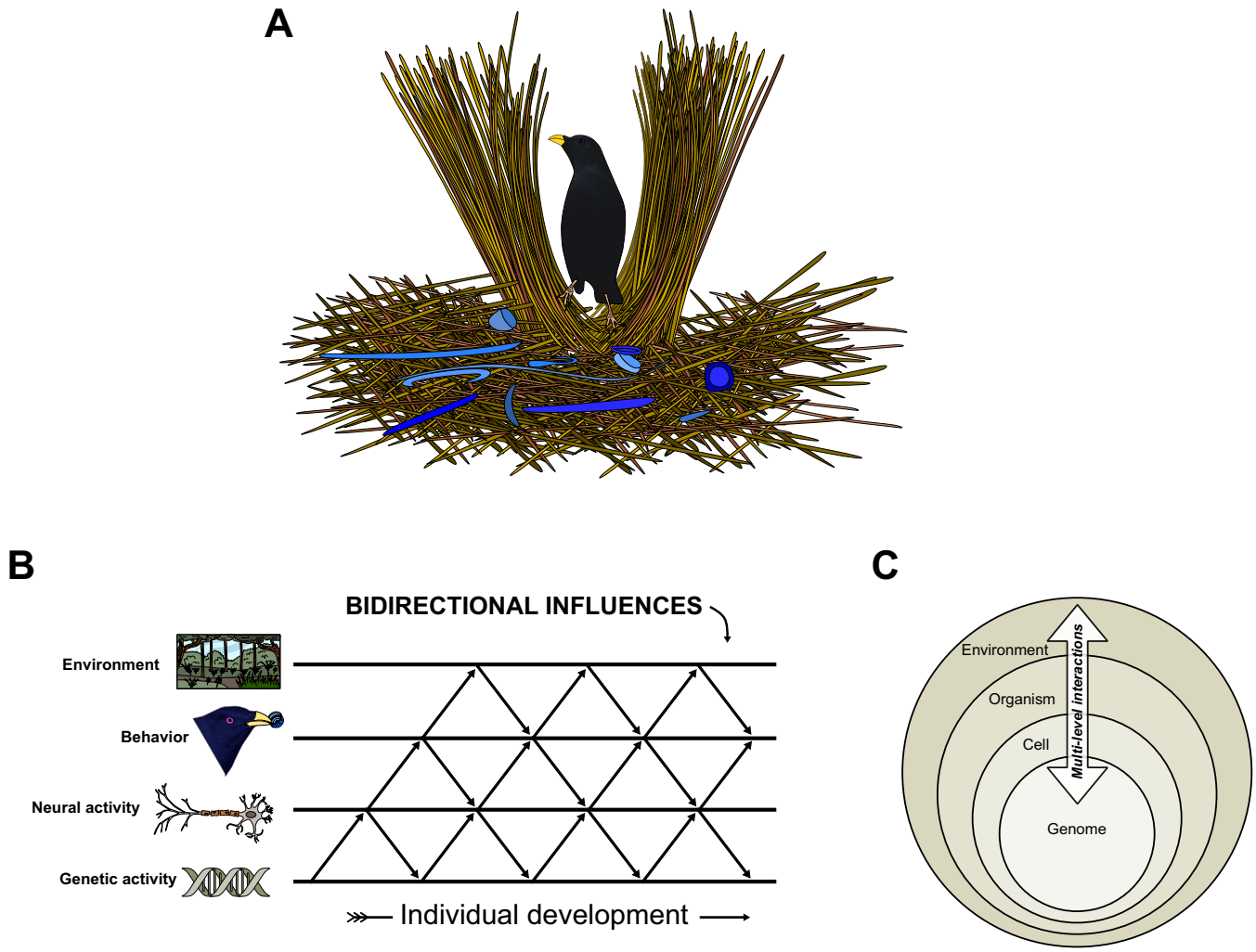


FIGURE 1 | (A) A male satin bowerbird in its bower. Drawing based on "Satin Bowerbird at Bower" by J. J. Harrison, used under CC BY-SA 4.0 via Wikimedia Commons (https://tinyurl.com/2u92vb9h). (B, C) Similarities between conceptions of ontogeny in developmental psychobiology and the EES. (B) Gottlieb's model of probabilistic epigenesis, modified from Gottlieb (2007). In Gottlieb's original scheme, the environment encompasses the physical, social, and cultural environment, but here only the physical environment is depicted. (C) EES' constructive development, as modified from Laland et al. (2015). Notice the equivalence between both ideas. The bowerbird in B was drawn based on "Satin Bowerbird JCB" by Joseph C. Boone, used under CC BY-SA 4.0 via Wikimedia Commons (https://tinyurl.com/y5vatcsd).

thus of limited relevance for explaining evolutionary change (Mayr 1961; Laland et al. 2011).

Although these views dominated for part of the twentieth century, once a perspective becomes established, alternatives tend to emerge. For the traditional ethological view, one alternative became known as developmental psychobiology (Michel and Moore 1995), which is characterized by the rejection of the nature vs. nurture dichotomy. For example, the concept of the 'ontogenetic niche' introduced by Meredith J. West and Andrew P. King emphasizes that behavioral phenotypes arise through dynamic interactions between genes, organisms, and the environment, with no privileged causal factor (West and King 1987). A similar response occurred in evolutionary biology, as researchers questioned the proximate-ultimate distinction and reconsidered development, plasticity, and behavior as causes, rather than just consequences, of evolution (Laland et al. 2011; Uller and Laland 2019; West-Eberhard 2003). This shift in explanatory standards is characteristic of the "extended evolutionary synthesis", a recent perspective in which the principle of constructive development acquires a central role, distinguishing it from the explanatory framework of the modern synthesis (Lala et al. 2024; Laland et al. 2015).

In this paper, we ask to what extent the notion of constructive development in evolutionary biology has been influenced, directly or indirectly, by the developmental psychobiology tradition in ethology, and by the work of Meredith J. West and Andrew P. King in particular. We first briefly review the intellectual history of this movement and the key characteristics of West and King's ontogenetic niche concept (West and King 1987), which stands out as their most influential theoretical contribution. We then outline contemporary development-centered approaches to evolution, highlighting their conceptual parallels with developmental psychobiology. Finally, we study how the ontogenetic niche concept made it to modern discussions about the role of development in evolution by conducting a targeted bibliometric analysis of the influence of West and King's ontogenetic niche paper (West and King 1987).

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2 | Developmental Psychobiology

Developmental psychobiology arose in response to the classic conception of instinct, which took shape in the wake of Darwin's theory of evolution and reached its peak with Konrad Lorenz's instinct theory (Beach 1955; Brigandt 2005; Lorenz and Tinbergen 1938; Richards 1987). Lorenz considered instincts innate, adaptive behavioral patterns that are mechanically elicited by specific environmental stimuli. As such, the theory presupposed a separation between nature and nurture, a bottom-up, unidirectional view of causation, and a limited role for development (Blumberg 2017; Brigandt 2005). Similar assumptions—yet from the "nurture" side—were made across several psychological disciplines and schools, including physiological and behavioristic psychology (Greenwood 2015).

The precursors of developmental psychobiology questioned these presuppositions. Zing-Yang Kuo (1898–1970) considered species-specific behaviors the product of interactions between the organism and the environment across development. He argued, for example, that the allegedly instinctive pecking behavior of the chick was influenced by early embryonic experiences, stressing the importance of the prenatal period (Hinde 1966; Kuo 1932; Richards 1987). Theodore C. Schneirla (1902–1968) questioned the idea of a privileged level of explanation. In his view, doryline ants' cyclic colony patterns resulted from multiple processes taking place across different levels of organization, "always (...) dependent upon conditions in the species-typical environment" (Schneirla 1966, 291; see also Kohn 2024). To this end, he conferred the terms of maturation and experience new meanings. These and other scholars, including Hebb (1953), Beach (1955), and Lehrman (1953), argued for the rejection of the instinct concept and the innate versus learned dichotomy, laying the conceptual ground for the emergence of developmental psychobiology.

Contrasting with traditional ethology, the developmental psychobiology tradition is characterized by the primacy of development, a bidirectional conception of causation, and an emphasis on the environment. First, for developmental psychobiologists, development has a privileged status both as a focus of research and a source of explanations. Second, in this tradition, causes are not assumed to flow bottom-up but are thought to occur bidirectionally across all levels of organization. Finally, the emphasis is on the environment (including the social environment) as an essential component of normal development—organisms expect certain kinds of environmental inputs, which become as indispensable as genes for the correct fruition of ontogeny. This made it different from research that focused on adverse effects of toxic environments (e.g., Cohlan 1953; Wilson 1959).

Developmental psychobiology has stimulated multiple research programs (see Michel 2013). For example, Gottlieb's work on the mallard duck (*Anas platyrhynchos*) confirmed the role of embryonic self-stimulation in the development of birds' recognition of conspecifics (Gottlieb 1991a). Moore's research on mother rats' licking behavior of male pups' anal-genital regions revealed that maternal stimulation is crucial for their normal adult sexual behavior (Michel 2013; Moore 1995). Similarly, Meaney's and others' investigations showed the importance of maternal care for the transgenerational transmission of stress responsiveness (Champagne and Curley 2009; Francis et al. 1999; Meaney 2001; Weaver et al. 2004). Empirical work in developmental psychobiology was also accompanied by a rich body of theoretical literature, including Gottlieb's own considerations about the experiential canalization of behavioral development (Gottlieb 1991b, 1997, 2007) or Donald Hebb's theory of synaptic plasticity (Hebb 1949). Even if Lorenz and others criticized the arguments and evidential basis of developmental psychobiology (e.g., Lorenz 1981), this approach eventually permeated conventional ethology, which came to adopt many of its postulates (Bateson and Laland 2013; Hinde 1966; Tinbergen 1963).

3 | West and King's Ontogenetic Niche

The emphasis on development and the importance of environmental and social factors come together in the concept of the ontogenetic niche, coined by Meredith J. West and Andrew P. King in 1987 (West and King 1987). Building on insights from child developmental psychology (e.g., Lerner and Busch-Rossnagel 1981) and their own research on bird song development (e.g., King et al. 1986; King and West 1983a, 1983b; West and King 1985), West and King defined the ontogenetic niche as the "ecological and social legacies that accompany genes" (p. 549). The outspoken aim was to move past the view of nature and nurture as contrasting factors with distinct relative explanatory weights, conceiving them "not as rivals, but as kin" (p. 559; see also West et al. 1990).

This kinship would manifest in three interrelated ways: niches as legacies, niches as links, and niches as ways of life. The "niche as legacy" emphasizes that "genes inherit a rich and supportive environment" (p. 552), with genes being inert molecules that do little unless placed into a rich somatic and environmental milieu, which is crucial for species-typical development. This legacy justifies behavioral researchers to "ask not what's inside the genes (...), but what the genes (...) are inside of" (p. 552). The "niche as link", in turn, poses that development and inheritance, at least in social animals like birds, "rests on transgenerational social dynamics" (p. 554). Organisms pass on many more resources than genes to the next generation; for example, the egg contains nutrients and molecules that act as templates in development, while parents build nests and connect offspring to resources that they will need later in life, such as territories and foraging sites (Uller 2012). These exogenetic forms of inheritance are vulnerable to alteration, so both generations should make efforts to preserve them. This establishes a connection: "parents and offspring are inextricably linked in the efforts to preserve the value of exogenetically transmitted properties" (p. 554). Finally, the ontogenetic niche also constitutes "a way of life". Niches specify "the behavioral adaptations of [their] occupants" (p. 556). Each developmental stage is at the same time an adaptation to the organism's immediate circumstances and a facilitator of new ontogenetic (and phylogenetic) stages. Thus, parents and offspring "change in ways that serve to advance both into new behavioral states" (p. 556).

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4 | The Extended Evolutionary Synthesis

West and King point out that the ontogenetic niche demands a developmental-ecological approach, examining ontogenetic and environmental roots of species-typical behavior. A similar standpoint has recently gained popularity in evolutionary biology. Proponents of the "extended evolutionary synthesis", as it is often referred to, argue that what organisms do and experience during their lifetimes matters for evolution, so evolutionary biologists must take development and organismal responses to their environments seriously (Lala et al. 2024; Laland et al. 2011, 2015). This applies not only to vertebrates, but also invertebrates (e.g., Schwab et al. 2017), plants (e.g., Sultan 2015), and other organisms (e.g., Davison et al. 2025).

The extended evolutionary synthesis (henceforth EES) represents a break with the traditional explanatory framework, the modern synthesis (MS), which has tended to downplay developmental processes (e.g., Amundson 2005). The MS's disregard for development stems from two meta-scientific principles (see Mayr 1961). The first is the genetic program. The genetic program posits that organisms arise from genetically encoded instructions, which provide blueprints for constructing phenotypes. The second is the proximate-ultimate distinction, which splits biological causes into two discrete classes. Proximate causes are those that result from the deployment of a genetic program, governing the inner workings of organisms. They include hormonal, physiological, and developmental mechanisms. Ultimate causes, in turn, are those encompassing the transgenerational population-level processes that introduce and sort among programs, including mutation, migration, natural selection, and genetic drift. According to the MS, proximate and ultimate causes are independent: proximate mechanisms cannot modify inherited genetic programs and, therefore, they do not constitute causes of evolution. According to this perspective, evolutionary biologists are justified in ignoring development to explain adaptive change.

The EES questions the MS meta-scientific principles, replacing them with two alternatives: constructive development and reciprocal causation (Laland et al. 2015). Constructive development denotes the capacity of organisms to "shape [their] own developmental trajectory by constantly responding to, and altering, internal and external states" (Laland et al. 2015, 6). This means that development is underdetermined by genes and that the fate of the organism depends on the interaction between different systems (genes, cells, tissues, and the environment). Moreover, many organismal systems show exploratory behavior, which allows for functional responses not encoded in the genome. Some of these functional responses might even be inherited, so constructive development opposes both the proximate-ultimate distinction and the genetic program (Badyaev and Uller 2009).

Reciprocal causation reconsiders the relationship between organisms and evolution. If development is constructive rather than programmed, it follows that organisms may be causes of evolution in addition to their products. Living beings are flexible agents whose goal-directed development and behavior can influence evolutionary dynamics (Walsh 2015). Reciprocal causation thus stands in opposition to the proximate-ultimate distinction, allowing development and behavior to contribute to the explanation of adaptive diversification (Laland et al. 2011).

An EES perspective motivates investigations of development, plasticity, and organism–environment feedback as causes, and not just mere consequences, of evolutionary change. For example, Duckworth's work on western bluebirds (*Sialia mexicana*) shows that maternal effects and behavioral plasticity can facilitate rapid adaptive divergence during range expansions (Duckworth et al. 2015). Research on the hermit crab *Coenobita compressus* suggests that the reliance on modified shells can bias selective pressures towards increased social tolerance (Laidre 2012, 2019), while a diverse body of research on animals and plants is dedicated to identifying when and why plasticity can initiate and direct adaptive evolution along the "lines of least developmental resistance" (Brun-Usan et al. 2021; Feiner et al. 2020; Levis et al. 2018; Picao-Osorio et al. 2025; Radersma et al. 2020; Tsuru and Furusawa 2025; for overviews, see Pfennig 2021; Uller et al. 2018, 2024).

5 | Connections Between Developmental Psychobiology and the EES

While developmental and organism-centered approaches have a long tradition in evolutionary biology (Baedke and Fábregas-Tejeda 2023), there is no doubt that there has been a growth of interest in development, plasticity, and exogenic inheritance in the past 20 years (e.g., Badyaev and Uller 2009; Bonduriansky and Day 2020; Jablonka and Lamb 2005; Lala et al. 2024; Laland et al. 2015; Levis and Pfennig 2016; Müller 2007; Odling-Smee et al. 2003; West-Eberhard 2003). Within this developmental turn, the EES attempts to synthesize related approaches and explicitly anchor them into alternative meta-scientific principles (Laland et al. 2015). This makes it evident that developmental psychobiology and the EES share core assumptions. Specifically, their similarities are particularly noticeable in their views of ontogeny, reflected in developmental psychobiology concepts such as Gottlieb's probabilistic epigenesis or West and King's ontogenetic niche, and the EES' principle of constructive development (Figure 1B,C).

How did researchers in ethology and evolutionary biology come to put forward so similar views about development? A plausible conjecture is that while recent ideas concerning the primacy of development in evolution initially emerged without awareness of developmental psychobiology, the EES movement eventually discovered and made use of some of its insights to support its notion of constructive development. A comprehensive analysis of the role of developmental psychobiology in the emergence of the EES is beyond the scope of this paper. Instead, here we focus on the specific contribution of West and King's ontogenetic niche paper (1987), as this is the main theoretical work from the developmental psychobiology tradition cited in the influential paper on the conceptual foundations of the EES (Laland et al. 2015; with more than 1900 citations by September 2026). Can we identify specific fields, intellectual traditions, and authors that favored the transmission of the ontogenetic niche concept into evolutionary biology?

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5.1 | Fields

To address these questions, we applied tools from computational history and philosophy of science (Ramsey and De Block 2022; see Appendix S1 for a full description of the procedure followed and the techniques employed). First, using field-level keywords and topic modeling on the publications that have cited West and King (1987), we identified fields and topics where the ontogenetic niche has been referenced (Figure 2). More specifically, fields were identified by counting the number of field-level keywords in a tokenized version of the title, abstract, journal title, and brief article notes, granting double weight to particularly diagnostic keywords. Topics were assigned using collapsed Gibbs-sampled latent Dirichlet allocation (LDA), where words are iteratively assigned to a predefined number of topics based on how strongly each word is linked with each topic and how common each topic is within a given document (Malaterre et al. 2022). The corpus used for topic modeling was roughly the same as with field assignment (only the journal title and tokens without conceptual content were removed).

The fields in which West and King (1987) has been most cited are psychology, biology, philosophy (mostly philosophy of science), and medicine, in that order (see also Figure S1). In psychology, the ontogenetic niche has been most referenced in discussions about the nature versus nurture dichotomy and developmental systems theory (see below). In biology, it has been particularly cited in discussions about the developmental approach to evolution, with the other topic being what we have called "developmental ethology", that is, empirical work on the development of species-typical behavior in the wild. Despite their empirical work on bird song learning (e.g., King et al. 1986; King and West 1983a, 1983b; West and King 1985), West and King (1987) has been little cited in empirical research on birds (only 11 references after manually inspecting the "developmental ecology" and "developmental ethology" topics in psychology and biology, respectively). In philosophy, it enters, again, debates surrounding developmental approaches to nature and nurture. Finally, in medicine, the ontogenetic niche has been more referenced in the context of the developmental origins of psychiatric disorders.

Figure 2A suggests an interesting possible route by which the ontogenetic niche entered evolutionary biology discussions. Biology exhibits two periods of moderate increase in citations (early 1990s to around 2005, and early 2010s to the present) separated by a virtually stationary period. Philosophy's first citation of West and King (1987), in turn, arrived only in 2004, more than 15 years after its publication. This was Jason Robert's book *Embryology, Epigenesis, and Evolution: Taking Development Seriously* (Robert 2004), which mentions the ontogenetic niche in the context of a non-genecentric view of development (chap. 5, p. 87). The following citations correspond to a series of publications by Robert, Karola Stotz, and Paul E. Griffiths (Griffiths 2008; Robert 2008; Stotz 2008; Stotz and Griffiths 2008).

Could it be that the arrival of philosophical discussions of West and King (1987) favored the incorporation of the ontogenetic niche concept into biological discussions about the role of development in evolution? If that were the case, we would expect an increase in citations to West and King (1987) in biology publications belonging to the topic of "developmental approaches to evolution" after the first citing philosophy contribution (2005 onwards), when compared to the previous time period (i.e., from its first citation to 2004). Figure 3 shows a pattern compatible with that hypothesis: from the first citation in 1988 to 2004, the year when the first philosophy contribution citing West and King (1987) came out, we see that only around 35% of biology publications citing West and King (1987) belong to the topic of developmental approaches to evolution. Conversely, after philosophers started to cite the ontogenetic niche, the pattern changes so that a majority of biological papers citing West and King (1987) do so in discussions about the role of development in evolution. It seems as if philosophers had changed the way in which West and King (1987) is read and used by biologists.

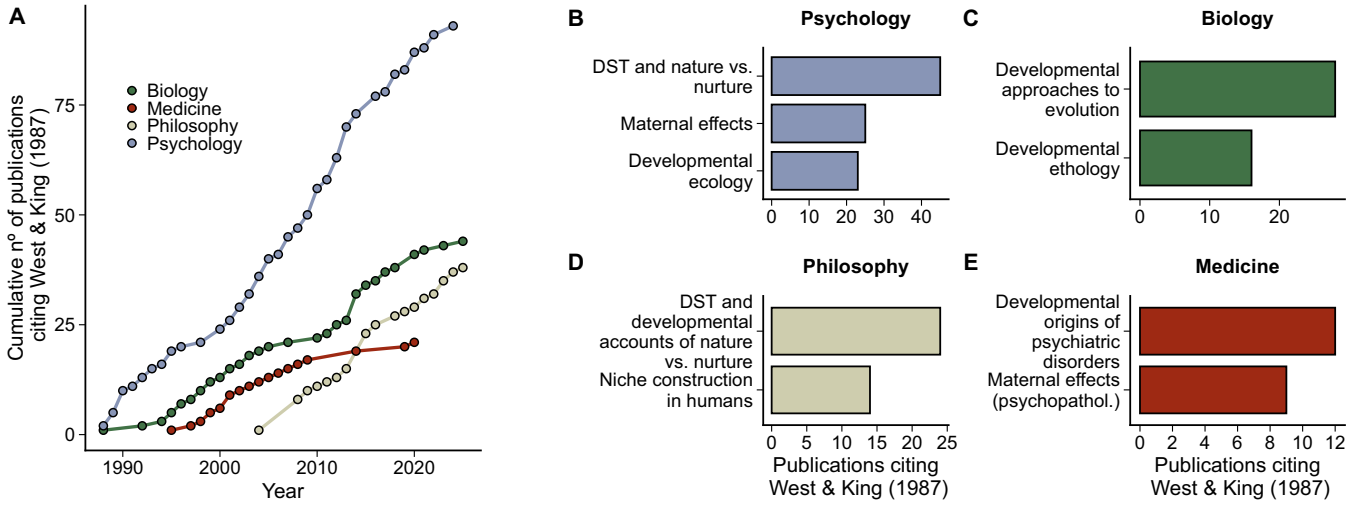


FIGURE 2 | (A) Cumulative number of citing publications per field over time since the publication of West and King (1987) ($N = 196$, excluding a publication that didn't fit any category). (B–E) Within each field, the topics discussed by the publications that have cited West and King (1987) (B: Psychology, C: Biology, D: Philosophy, E: Medicine). For a full description of each of these topics, see Appendix S1, section on *Topic determination* and Table S2. Note that the x-axes have different scales across panels (B)–(E), as the absolute number of citations of West and King (1987) varies by field.

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5.2 | Intellectual Traditions

Guided by this conjecture, we identified intellectual traditions within the philosophy of science that could have helped bring the ontogenetic niche to the developmental evolutionary biology literature (which includes the EES publications). We understand an intellectual tradition as a series of scholars who espouse a similar account of a concept—in this case, development. If a work (here, West and King 1987) is cited alongside publications of a well-known intellectual tradition, this might make such a work easier to digest for scholars unfamiliar with it, provided that these scholars are acquainted with the intellectual tradition with which it is co-cited. In that sense, if the ontogenetic niche concept had been paired with works of an intellectual tradition that is known and respected among evolutionary biologists, this might have facilitated its incorporation into discussions about a development-centered view of evolution.

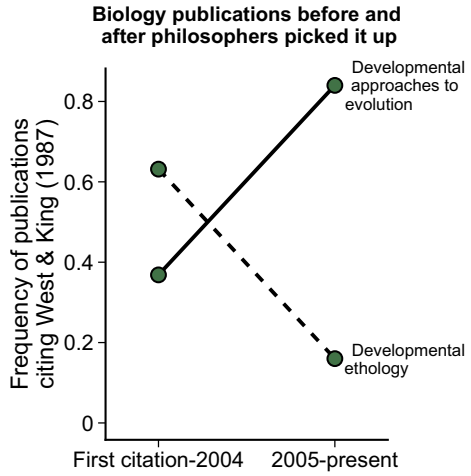


FIGURE 3 | Frequency of publications of the different biological topics in which West and King (1987) has been cited before (left) and after (right) philosophers started to discuss this work ($N = 44$ for all periods and topics).

We sought these traditions using the complete reference list of each of the 186 works that cite West and King (1987); after removing 11 works without DOIs from the original 197 publications; see Appendix S1. Specifically, we explored which publications had been cited more frequently alongside West and King (1987), looking for common threads between them (i.e., a co-citation analysis; Zupic and Čater 2015). These include Lehrman's classic critique of Lorenz's instinct theory (Lehrman 1953; co-cited in 41 publications), Susan Oyama's book *The Ontogeny of Information* (Oyama 1986; co-cited in 33), Paul E. Griffiths and Russell D. Gray's article on *Developmental Systems and Evolutionary Explanation* (Griffiths and Gray 1994; co-cited in 33 works), Eva Jablonka and Marion Lamb's book *Evolution in Four Dimensions* (Jablonka and Lamb 2005; co-cited 32 times), and Susan Oyama, Paul E. Griffiths, and Russell D. Gray's volume *Cycles of Contingency* (Oyama et al. 2001; co-cited in 30 publications) (Figure 4).

As the topics across fields, the co-citation patterns suggest a key role for the intellectual tradition of developmental systems theory (DST). DST is a "general theoretical perspective on development, heredity, and evolution" that is centered around the idea of a "developmental system", or the "matrix of resources that are needed to reproduce the life cycle" (Griffiths and Hochman 2015, 1; Oyama et al. 2001, 2). According to DST, it is these resources that are inherited and respond to natural selection, and therefore, that should constitute the focus of evolutionary biology. Oyama (1986), Griffiths and Gray (1994), and Oyama et al. (2001), which feature prominently in the co-citation counts, constitute key references within such a tradition.

DST is particularly well-positioned to provide an understanding of development and the ontogenetic niche that resonates across fields, as the tradition is well known among both psychologists and biologists. As for biology, many DST publications feature or discuss works by Richard Lewontin, Conrad H. Waddington, and other forerunners of the organismal approach. Likewise, developmental psychobiologists have often contributed to DST publications, either directly or through reprints, including Gilbert Gottlieb, Timothy Johnston, and Daniel Lehrman

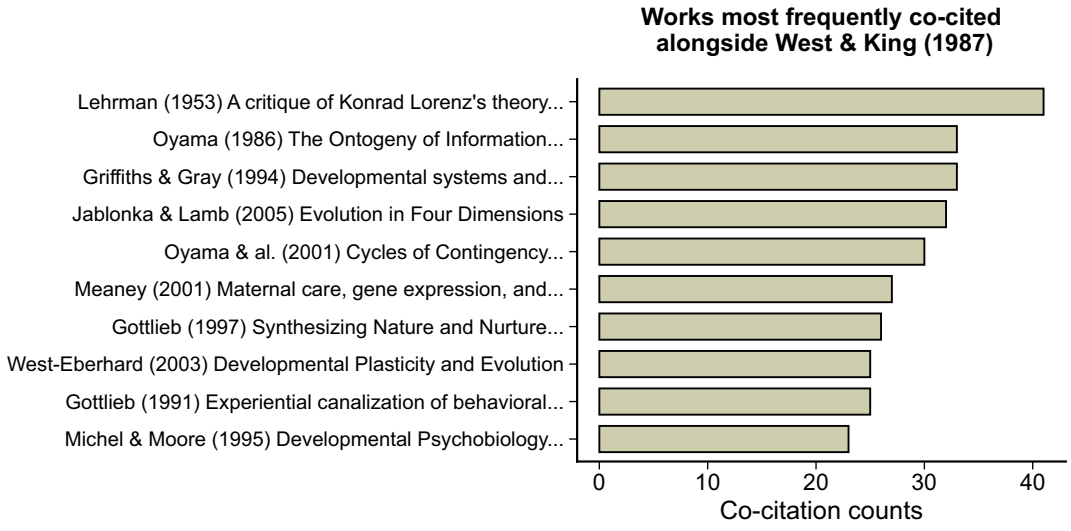


FIGURE 4 | The 10 works that have been most frequently co-cited alongside West and King (1987) in the works that have cited this publication ($N = 186$ out of the original 197 works; see Appendix S1 for details).

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(Oyama et al. 2001). Even the very label "DST" was itself invented by psychologists (Ford and Lerner 1992; Griffiths and Hochman 2015), although today the tradition is perhaps mostly associated with the philosophy of evolutionary biology.

Whether DST works are well-connected to the literature in both camps was explored with a co-citation network (Newman 2010; see a detailed explanation and the full result details in Appendix S1). Co-citation networks explore co-citation patterns by representing publications as nodes and the number of times two given works have been cited together as links or edges. In this case, the nodes of the network were the 100 works with which West and King (1987) has most frequently been co-cited, while the links represented the number of publications in which these works have been cited together (Figure 5). This allowed us to determine the connections that exist between the works that are most prominently co-cited with the ontogenetic niche paper, exploring the centrality and distribution of DST publications within such works.

The community detection technique used was Clauset-Newman-Moore fast greedy modularity optimization algorithm (Barabási 2016). This algorithm initially assigns nodes to self-contained communities, and then iteratively merges neighboring communities whenever this move increases the total modularity of the network. The Clauset-Newman-Moore algorithm identified three sub-optimally partitioned communities within the network ($M = 0.178$). Community #1 aggregates publications in developmental psychobiology (including Lehrman 1953, various works by Gottlieb, and many publications on maternal effects), as well as some works by philosophers of biology (Keller 2000; Moss 2002). Community #2 mostly clumps works in developmentally-oriented evolutionary biology, including West-Eberhard (2003) and Laland et al.'s (2015) EES programmatic paper. Additionally, it includes some publications in the philosophy of biology and DST (e.g., Griffiths and Gray 1994; Oyama et al. 2001). Finally, community #3 mostly combines further works in DST and developmental psychobiology (including publications by Oyama, Gottlieb, or Kuo). This suggests that the conceptual landscape of the ontogenetic niche, composed of the works most frequently associated with West and King (1987), is structured around organismal evolutionary biology, developmental psychobiology, and DST-philosophy of biology—and that of these, the latter is particularly spread across

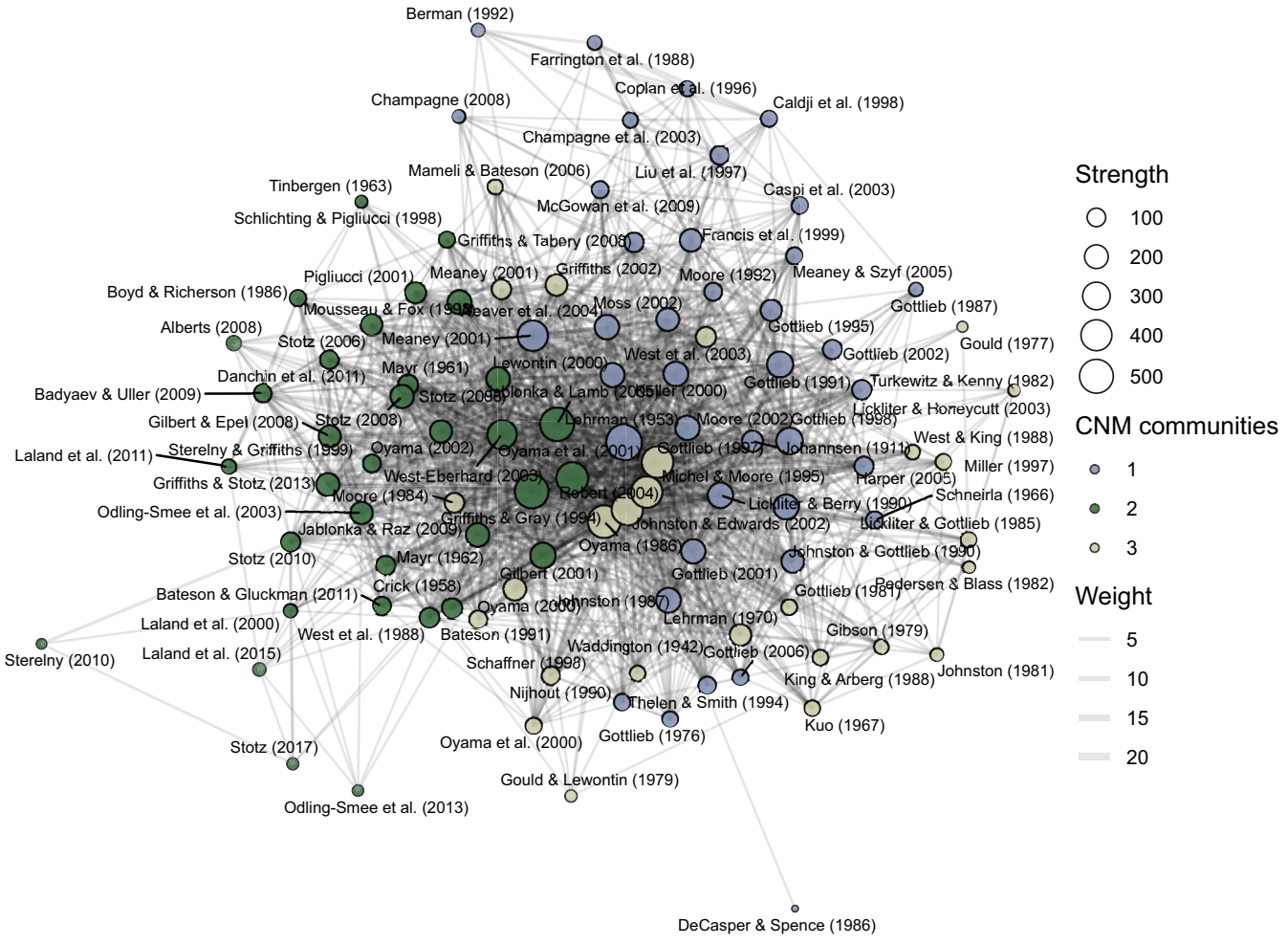


FIGURE 5 | Network of works surrounding West and King (1987). Nodes represent the 100 works most frequently cited alongside this paper. Links denote the weight of their connections, that is, in how many publications they co-occur. Node size represents each node's strength, or the sum of weights of all links connected to a node. Layout uses stress majorization, which brings closely connected nodes spatially closer. Finally, the colors represent the community to which the different nodes belong, according to the Clauset-Newman-Moore fast greedy modularity optimization algorithm (see Appendix S1 for a detailed explanation of the procedure followed). Given that the communities somehow represent different fields, they have been colored following the scales used in Figure 2B–E. Note, however, that they were obtained using different methodologies.

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communities. This is compatible with the idea that philosophy and DST have acted as bridges between camps, a claim that is further suggested by the works with higher betweenness centrality and strength in the network (see Figure S3).

5.3 | Authors

Finally, are there particular authors who explain the above network structure? That is, scholars who, by fitting the ontogenetic niche within DST, could have helped developmentally-oriented evolutionary biologists learn about the concept and cite it? To answer this, we asked: which scholars are more likely to import a concept from one field into another? Two factors might seem a priori most likely to help act as a "bridge". First, it is important that an author habitually mentions the concept in their work (or that they cite its main associated reference often). Second, it also matters how interdisciplinary such an author is; that is, that they publish evenly across fields. This was explored by measuring the interdisciplinarity of the authors that cite West and King (1987) more often (Figure 6). To account for co-authorship inflation, each author's citation of West and King (1987) was fractioned by the total number of authors in the publication (i.e., 1/*number of authors*). The citing authors' interdisciplinarity was measured using the Shannon index (H') over the fields to which their publications citing West and King (1987) belong (see Appendix S1 for a complete description of the analyses).

This analysis revealed philosophers of science Paul E. Griffiths and Karola Stotz, as well as psychologist Robert Lickliter, as the best-positioned authors for having helped transfer the ontogenetic niche concept from developmental psychobiology to organismal evolutionary biology (similar results are obtained if a different measure of interdisciplinarity, which we have called "semantic interdisciplinarity", is used; see Figure S5). Of these authors, Griffiths and Stotz stand as particularly likely candidates for having carried out the task, since, for instance, Griffiths and Stotz's book *Genetics and Philosophy: An Introduction* (Griffiths and Stotz, 2013) is mentioned several times in Laland et al.'s (2015) EES paper. This very same book references, in turn, West and King (1987), which is cited a total of seven times (pp. 5, 110, 112, 129, 134). Stotz and Griffiths also reference several other works by West and King, mostly in discussions of topics of concern to evolutionary biologists (e.g., heredity). This suggests it was primarily these two philosophers who fitted the ontogenetic niche into a DST framework from which developmentally-oriented evolutionary biologists could pick it up and import it into their own work.

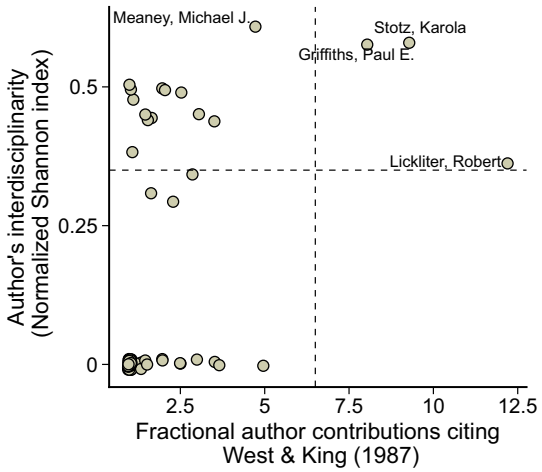


FIGURE 6 | Authors' interdisciplinarity (normalized Shannon index over the fields to which their contributions belong) and weighted intensity of West and King (1987) citation (fractional contributions citing this work, to account for co-authorship; see Appendix S1 for a detailed explanation). Each data point constitutes an author. Only authors with equal or more than 1 fractional contribution are displayed. This corresponds, for instance, to authors with 2 publications if each of these has two authors, or 3 publications if the publication has three authors. The quadrants help visualize the most interdisciplinary authors who have cited West and King (1987) the most (upper right quadrant).

6 | Concluding Remarks

The ontogenetic niche concept was originally developed by West and King (1987) to emphasize how the recurrent expression of phenotypes across generations relies on environmental, social, and organismal factors. Using tools from computational history and philosophy of science, we have identified potential fields, intellectual traditions, and authors that could have facilitated the transfer of this conceptual framework into evolutionary biology. The analyses suggest that this transfer occurred not directly from developmental psychobiology publications but indirectly through the field of philosophy of science. One philosophical tradition in particular, DST, could have been instrumental in that task. More specifically, we have argued that it was DST philosophers Paul E. Griffiths and Karola Stotz who placed the ontogenetic niche concept in a theoretical context that was understandable and relevant to developmentally minded evolutionary biologists, among which it ultimately became well-received.

What can we infer from the above results? Being focused on just one work (i.e., West and King 1987), our inferences are necessarily preliminary. For example, we cannot rule out that the results of Figure 3 are partly due to the general increase of interest in developmental approaches to evolution in the last two decades. Future studies should analyze this in more detail. Nevertheless, our study points to a plausible hypothesis, namely, that philosophers of science acted as bridges between similar ways of thinking in separate scientific fields. This hypothesis would be corroborated if philosophical publications belonging to the DST tradition also feature in other instances of cross-citation between developmental psychobiology and developmentally-oriented evolutionary biology. More generally, the bridge-building role of philosophers could also be tested in similar scenarios, exploring, for example, whether contemporary citations of classical discussions about the relationship between evolution, behavior, and development (e.g., James M. Baldwin's *Development and Evolution*; Baldwin 1902) are also mediated by philosophers (i.e., whether modern philosophical publications are cited alongside classical references, providing them with a context).

Philosophy-mediated cross-disciplinary interaction can be important since scientific investigation is ever more specialized. The growing sophistication of data collection techniques and conceptual frameworks makes it increasingly difficult for scientists to venture outside their domains of expertise without being penalized

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(e.g., Yanai and Lercher 2020). While specialization remains a necessary condition for the progress of science (Kuhn 1970), it can also have drawbacks. Particularly, it might hinder innovation, which often results from the combination of hitherto separated ideas (Wang and Barabási 2021), a process for which diving into fields outside one's scientific specialty might help.

Despite the increasing "scientization" of philosophy, philosophers of science still retain the habit of moving freely across disciplines. This habit is foundational; philosophy has traditionally sought to provide the "big picture", for which transcending disciplinary divides is necessary. This often results in innovative theoretical proposals, such as DST in evolutionary biology or Patricia Churchland's neurophilosophical project in psychology and neuroscience (Churchland 1986). An indirect way of fostering productive cross-disciplinary exchange is therefore for scientists to be open to philosophy contributions. There is an increasing trend of philosophers being involved in resolving scientific problems (e.g., Love and Wagner 2022; as well as chapters within Hansen et al. 2023 and Uller and Laland 2019). This requires open mindedness and willingness to engage with different disciplinary conventions. The transfer of concepts and ideas from developmental psychobiology into evolutionary biology is an example of how this effort can pay off.

Author Contributions

Juan Gefaell: conceptualization, methodology, investigation, data curation, formal analysis, visualization, resources, writing – original draft, writing – review and editing. **Tobias Uller:** conceptualization, methodology, funding acquisition, writing – original draft, writing – review and editing, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in WestKingImpact at https://github.com/juangefaell/WestKingImpact.

References

Amundson, R. 2005. *The Changing Role of the Embryo in Evolutionary Thought: Roots of Evo-Devo*. Cambridge University Press. https://doi.org/10.1017/CBO9781139164856.

Badyaev, A. V., and T. Uller. 2009. "Parental Effects in Ecology and Evolution: Mechanisms, Processes and Implications." *Philosophical Transactions of the Royal Society, B: Biological Sciences* 364, no. 1520: 1169–1177. https://doi.org/10.1098/rstb.2008.0302.

Baedke, J., and A. Fábregas-Tejeda. 2023. "The Organism in Evolutionary Explanation: From Early Twentieth Century to the Extended Evolutionary Synthesis." In *Evolutionary Biology: Contemporary and Historical Reflections Upon Core Theory*, edited by T. E. Dickins and B. J. A. Dickins, 121–150. Springer International Publishing. https://doi.org/10.1007/978-3-031-22028-9_8.

Baldwin, J. M. 1902. *Development and Evolution*. Macmillan.

Barabási, A.-L. 2016. *Network Science*. Cambridge University Press.

Bateson, P., and K. N. Laland. 2013. "Tinbergen's Four Questions: An Appreciation and an Update." *Trends in Ecology & Evolution* 28, no. 12: 712–718. https://doi.org/10.1016/j.tree.2013.09.013.

Beach, F. A. 1955. "The Descent of Instinct." *Psychological Review* 62, no. 6: 401–410. https://doi.org/10.1037/h0049138.

Blumberg, M. S. 2017. "Development Evolving: The Origins and Meanings of Instinct." *WIREs Cognitive Science* 8, no. 1–2: e1371. https://doi.org/10.1002/wcs.1371.

Bonduriansky, R., and T. Day. 2020. *Extended Heredity: A New Understanding of Inheritance and Evolution*. Princeton University Press.

Borgia, G. 1985. "Bower Quality, Number of Decorations and Mating Success of Male Satin Bowerbirds (*Ptilonorhynchus violaceus*): An Experimental Analysis." *Animal Behaviour* 33, no. 1: 266–271. https://doi.org/10.1016/S0003-3472(85)80140-8.

Brigandt, I. 2005. "The Instinct Concept of the Early Konrad Lorenz." *Journal of the History of Biology* 38, no. 3: 571–608. https://doi.org/10.1007/s10739-005-6544-3.

Brun-Usan, M., A. Rago, C. Thies, T. Uller, and R. A. Watson. 2021. "Development and Selective Grain Make Plasticity "Take the Lead" in Adaptive Evolution." *BMC Ecology and Evolution* 21, no. 1: 205. https://doi.org/10.1186/s12862-021-01936-0.

Champagne, F. A., and J. P. Curley. 2009. "Epigenetic Mechanisms Mediating the Long-Term Effects of Maternal Care on Development." *Neuroscience & Biobehavioral Reviews* 33: 593–600. https://doi.org/10.1016/j.neubiorev.2007.10.009.

Churchland, P. S. 1986. *Neurophilosophy: Toward a Unified Science of the Mind-Brain*. MIT Press. https://doi.org/10.7551/mitpress/4952.001.0001.

Cohlan, S. Q. 1953. "Excessive Intake of Vitamin A as a Cause of Congenital Anomalies in the Rat." *Science* 117, no. 3046: 535–536. https://doi.org/10.1126/science.117.3046.535.

Davison, D. R., A. M. Nedelcu, O. D. A. Eneji, and R. E. Michod. 2025. "Plasticity and the Evolution of Group-Level Regulation of Cellular Differentiation in the Volvocine Algae." *Proceedings of the Royal Society B: Biological Sciences* 292, no. 2043: 20242477. https://doi.org/10.1098/rspb.2024.2477.

Duckworth, R. A., V. Belloni, and S. R. Anderson. 2015. "Cycles of Species Replacement Emerge From Locally Induced Maternal Effects on Offspring Behavior in a Passerine Bird." *Science* 347, no. 6224: 875–877. https://doi.org/10.1126/science.1260154.

Feiner, N., I. S. Jackson, K. L. Munch, R. Radersma, and T. Uller. 2020. "Plasticity and Evolutionary Convergence in the Locomotor Skeleton of Greater Antillean *Anolis* Lizards." *eLife* 9: e57468. https://doi.org/10.7554/eLife.57468.

Ford, D. H., and R. M. Lerner. 1992. *Developmental Systems Theory: An Integrative Approach*. SAGE Publications.

Francis, D., J. Diorio, D. Liu, and M. J. Meaney. 1999. "Nongenomic Transmission Across Generations of Maternal Behavior and Stress

14390310, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/eth.70125 by Juan Gefaell - Universidad de Vigo , Wiley Online Library on [29/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Responses in the Rat." *Science* 286, no. 5442: 1155–1158. https://doi.org/10.1126/science.286.5442.1155.

Gottlieb, G. 1991a. "Experiential Canalization of Behavioral Development: Results." *Developmental Psychology* 27, no. 1: 35–39. https://doi.org/10.1037/0012-1649.27.1.35.

Gottlieb, G. 1991b. "Experiential Canalization of Behavioral Development: Theory." *Developmental Psychology* 27, no. 1: 4–13. https://doi.org/10.1037/0012-1649.27.1.4.

Gottlieb, G. 1997. *Synthesizing Nature-Nurture: Prenatal Roots of Instinctive Behavior*. Psychology Press. https://doi.org/10.4324/9781315805788.

Gottlieb, G. 2007. "Probabilistic epigenesis." *Developmental Science* 10, no. 1: 1–11. https://doi.org/10.1111/j.1467-7687.2007.00556.x.

Greenwood, J. D. 2015. *A Conceptual History of Psychology: Exploring the Tangled Web*. Cambridge University Press. https://doi.org/10.1017/CBO9781107414914.

Griffiths, P., and K. Stotz. 2013. *Genetics and Philosophy: An Introduction*. Cambridge University Press. https://doi.org/10.1017/CBO9780511744082.

Griffiths, P. E. 2008. "Jesse Prinz Gut Reactions: A Perceptual Theory of Emotion." *British Journal for the Philosophy of Science* 59, no. 3: 559–567. https://doi.org/10.1093/bjps/axn020.

Griffiths, P. E., and R. D. Gray. 1994. "Developmental Systems and Evolutionary Explanation." *Journal of Philosophy* 91, no. 6: 277–304. https://doi.org/10.2307/2940982.

Griffiths, P. E., and A. Hochman. 2015. "Developmental Systems Theory." In *Encyclopedia of Life Sciences*, 1–7. John Wiley & Sons, Ltd. https://doi.org/10.1002/9780470015902.a0003452.pub2.

Hansen, T. F., D. Houle, M. Pavličev, and C. Pélabon, eds. 2023. *Evolvability: A Unifying Concept in Evolutionary Biology?* MIT Press. https://doi.org/10.7551/mitpress/14126.001.0001.

Hebb, D. 1949. *The Organization of Behavior: A Neuropsychological Theory*. Wiley & Sons.

Hebb, D. O. 1953. "Heredity and Environment in Mammalian Behaviour." *British Journal of Animal Behaviour* 1: 43–47. https://doi.org/10.1016/S0950-5601(53)80053-5.

Hinde, R. A. 1966. *Animal Behavior: A Synthesis of Ethology and Comparative Psychology*. McGraw-Hill.

Jablonka, E., and M. J. Lamb. 2005. *Evolution in Four Dimensions*. MIT Press.

Keller, E. F. 2000. *The Century of the Gene*. Harvard University Press. https://doi.org/10.4159/9780674039438.

King, A. P., and M. J. West. 1983a. "Dissecting Cowbird Song Potency: Assessing a Song's Geographic Identity and Relative Appeal." *Zeitschrift für Tierpsychologie* 63, no. 1: 37–50. https://doi.org/10.1111/j.1439-0310.1983.tb00739.x.

King, A. P., and M. J. West. 1983b. "Epigenesis of Cowbird Song—A Joint Endeavour of Males and Females." *Nature* 305, no. 5936: 704–706. https://doi.org/10.1038/305704a0.

King, A. P., M. J. West, and D. H. Eastzer. 1986. "Female Cowbird Song Perception: Evidence for Different Developmental Programs Within the Same Subspecies." *Ethology* 72, no. 2: 89–98. https://doi.org/10.1111/j.1439-0310.1986.tb00609.x.

Kohn, G. M. 2024. "Revisiting T. C. Schneirla's "Interrelationships of the 'Innate' and the 'Acquired' in Instinctive Behavior" (1956)." *Biological Theory* 19, no. 2: 84–93. https://doi.org/10.1007/s13752-024-00458-4.

Kuhn, T. S. 1970. *The Structure of Scientific Revolutions*. 2nd, Enlarged ed. University of Chicago Press.

Kuo, Z. Y. 1932. "Ontogeny of Embryonic Behavior in Aves. IV. The Influence of Embryonic Movements Upon the Behavior After Hatching." *Journal of Comparative Psychology* 14, no. 1: 109–122. https://doi.org/10.1037/h0071451.

Laidre, M. E. 2012. "Niche Construction Drives Social Dependence in Hermit Crabs." *Current Biology* 22, no. 20: R861–R863. https://doi.org/10.1016/j.cub.2012.08.056.

Laidre, M. E. 2019. "Architectural Modification of Shells by Terrestrial Hermit Crabs Alters Social Dynamics in Later Generations." *Ecology* 100, no. 9: e02767. https://doi.org/10.1002/ecy.2767.

Lala, K. N., T. Uller, N. Feiner, M. W. Feldman, and S. F. Gilbert. 2024. *Evolution Evolving: The Developmental Origins of Adaptation and Biodiversity*. Princeton University Press. https://doi.org/10.2307/jj.13359158.

Laland, K. N., K. Sterelny, J. Odling-Smee, W. Hoppitt, and T. Uller. 2011. "Cause and Effect in Biology Revisited: Is Mayr's Proximate-Ultimate Dichotomy Still Useful?" *Science* 334, no. 6062: 1512–1516. https://doi.org/10.1126/science.1210879.

Laland, K. N., T. Uller, M. W. Feldman, et al. 2015. "The Extended Evolutionary Synthesis: Its Structure, Assumptions and Predictions." *Proceedings of the Royal Society B: Biological Sciences* 282, no. 1813: 20151019. https://doi.org/10.1098/rspb.2015.1019.

Lehrman, D. S. 1953. "A Critique of Konrad Lorenz's Theory of Instinctive Behavior." *Quarterly Review of Biology* 28, no. 4: 337–363. https://doi.org/10.1086/399858.

Lerner, R. M., and N. A. Busch-Rossnagel. 1981. *Individuals as Producers of Their Development. A Life-Span Perspective*. Academic Press.

Levis, N. A., A. J. Isdaner, and D. W. Pfennig. 2018. "Morphological Novelty Emerges From Pre-Existing Phenotypic Plasticity." *Nature Ecology & Evolution* 2, no. 8: 1289–1297. https://doi.org/10.1038/s41559-018-0601-8.

Levis, N. A., and D. W. Pfennig. 2016. "Evaluating 'Plasticity-First' Evolution in Nature: Key Criteria and Empirical Approaches." *Trends in Ecology & Evolution* 31, no. 7: 563–574. https://doi.org/10.1016/j.tree.2016.03.012.

Lorenz, K. 1981. *The Foundations of Ethology*. Springer.

Lorenz, K., and N. Tinbergen. 1938. "Taxis Und Instinkthandlung in der Eirollbewegung der Graugans. [Directed and Instinctive Behavior in the Egg Rolling Movements of the Gray Goose.]." *Zeitschrift für Tierpsychologie* 2: 1–29. https://doi.org/10.1111/j.1439-0310.1939.tb01558.x.

Love, A. C., and G. P. Wagner. 2022. "Co-Option of Stress Mechanisms in the Origin of Evolutionary Novelties." *Evolution* 76, no. 3: 394–413. https://doi.org/10.1111/evo.14421.

Malaterre, C., J. F. Chartier, and D. Pulizzotto. 2022. "Topic Modeling in HPS: Investigating Engaged Philosophy of Science Throughout the Twentieth Century." In *The Dynamics of Science: Computational Frontiers in History and Philosophy of Science*, edited by G. Ramsey and A. De Block, 164–185. University of Pittsburgh Press.

Mayr, E. 1961. "Cause and Effect in Biology." *Science* 134, no. 3489: 1501–1506. https://doi.org/10.1126/science.134.3489.1501.

Meaney, M. J. 2001. "Maternal Care, Gene Expression, and the Transmission of Individual Differences in Stress Reactivity Across Generations." *Annual Review of Neuroscience* 24: 1161–1192. https://doi.org/10.1146/annurev.neuro.24.1.1161.

Michel, G. F. 2013. "The Role of Developmental Psychobiology in the Unification of Psychology." *Review of General Psychology* 17, no. 2: 210–215. https://doi.org/10.1037/a0032936.

Michel, G. F., and C. L. Moore. 1995. *Developmental Psychobiology: An Interdisciplinary Science*. MIT Press.

Moore, C. L. 1995. "Maternal Contributions to Mammalian Reproductive Development and the Divergence of Males and Females." In *Advances in*

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the Study of Behavior, vol. 24, 47–118. Academic Press. https://doi.org/10.1016/S0065-3454(08)60392-7.

Moss, L. 2002. *What Genes Can't Do*. MIT.

Müller, G. B. 2007. "Evo–Devo: Extending the Evolutionary Synthesis." *Nature Reviews Genetics* 8, no. 12: 12. https://doi.org/10.1038/nrg2219.

Newman, M. E. J. 2010. *Networks: An Introduction*. Oxford University Press.

Odling-Smee, J., K. N. Lala, and M. Feldman. 2003. *Niche Construction: The Neglected Process in Evolution*. Princeton University Press.

Oyama, S. 1986. *The Ontogeny of Information: Developmental Systems and Evolution*. Duke University Press. https://doi.org/10.1215/9780822380665.

Oyama, S., R. D. Gray, and P. E. Griffiths. 2001. *Cycles of Contingency: Developmental Systems and Evolution*. MIT Press.

Pfennig, D. W. 2021. *Phenotypic Plasticity & Evolution: Causes, Consequences, Controversies*. CRC Press. https://www.routledge.com/Phenotypic-Plasticity--Evolution-Causes-Consequences-Controversies/Pfennig/p/book/9780367676551.

Picao-Osorio, J., C. Bouleau, P. M. Gonzalez de la Rosa, et al. 2025. "Evolution of Developmental Bias Explains Divergent Patterns of Phenotypic Evolution in Two Nematode Clades." *Proceedings of the National Academy of Sciences of the United States of America* 122, no. 34: e2507529122. https://doi.org/10.1073/pnas.2507529122.

Radersma, R., D. W. A. Noble, and T. Uller. 2020. "Plasticity Leaves a Phenotypic Signature During Local Adaptation." *Evolution Letters* 4, no. 4: 360–370. https://doi.org/10.1002/evl3.185.

Ramsey, G., and A. De Block. 2022. *The Dynamics of Science: Computational Frontiers in History and Philosophy of Science*. University of Pittsburgh Press. https://doi.org/10.2307/j.ctv31djr2f.

Richards, R. J. 1987. *Darwin and the Emergence of Evolutionary Theories of Mind and Behavior*. University of Chicago Press.

Robert, J. S. 2004. *Embryology, Epigenesis and Evolution: Taking Development Seriously*. Cambridge University Press. https://doi.org/10.1017/CBO9780511498541.

Robert, J. S. 2008. "The Comparative Biology of Human Nature." *Philosophical Psychology* 21, no. 3: 425–436. https://doi.org/10.1080/09515080802201237.

Schneirla, T. C. 1966. "Behavioral Development and Comparative Psychology." *Quarterly Review of Biology* 41, no. 3: 283–302. https://doi.org/10.1086/405056.

Schwab, D. B., S. Casasa, and A. P. Moczek. 2017. "Evidence of Developmental Niche Construction in Dung Beetles: Effects on Growth, Scaling, and Reproductive Success." *Ecology Letters* 20, no. 11: 1353–1363. https://doi.org/10.1111/ele.12830.

Stotz, K. 2008. "The Ingredients for a Postgenomic Synthesis of Nature and Nurture." *Philosophical Psychology* 21, no. 3: 359–381. https://doi.org/10.1080/09515080802200981.

Stotz, K., and P. E. Griffiths. 2008. "Biohumanities: Rethinking the Relationship Between Biosciences, Philosophy and History of Science, and Society." *Quarterly Review of Biology* 83, no. 1: 37–45. https://doi.org/10.1086/529561.

Sultan, S. E. 2015. *Organism and Environment: Ecological Development, Niche Construction, and Adaptation*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199587070.001.0001.

Tinbergen, N. 1951. *The Study of Instinct*. Oxford University Press.

Tinbergen, N. 1963. "On Aims and Methods of Ethology." *Zeitschrift für Tierpsychologie* 20, no. 4: 410–433. https://doi.org/10.1111/j.1439-0310.1963.tb01161.x.

Tsuru, S., and C. Furusawa. 2025. "Genetic Properties Underlying Transcriptional Variability Across Different Perturbations." *Nature Communications* 16: 2421. https://doi.org/10.1038/s41467-025-57642-8.

Uller, T. 2012. "Parental Effects in Development and Evolution." In *The Evolution of Parental Care*, edited by N. J. Royle, P. T. Smiseth, and M. Kölliker, 247–266. Oxford University Press.

Uller, T., and K. N. Laland, eds. 2019. *Evolutionary Causation: Biological and Philosophical Reflections*. MIT Press. https://doi.org/10.7551/mitpress/11693.001.0001.

Uller, T., L. Milocco, J. Isanta-Navarro, C. K. Cornwallis, and N. Feiner. 2024. "Twenty Years on From Developmental Plasticity and Evolution: Middle-Range Theories and How to Test Them." *Journal of Experimental Biology* 227, no. Suppl_1: jeb246375. https://doi.org/10.1242/jeb.246375.

Uller, T., A. P. Moczek, R. A. Watson, P. M. Brakefield, and K. N. Laland. 2018. "Developmental Bias and Evolution: A Regulatory Network Perspective." *Genetics* 209, no. 4: 949–966. https://doi.org/10.1534/genetics.118.300995.

Walsh, D. M. 2015. *Organisms, Agency, and Evolution*. Cambridge University Press. https://doi.org/10.1017/CBO9781316402719.

Wang, D., and A.-L. Barabási. 2021. *The Science of Science*. Cambridge University Press. https://doi.org/10.1017/9781108610834.

Weaver, I. C. G., N. Cervoni, F. A. Champagne, et al. 2004. "Epigenetic Programming by Maternal Behavior." *Nature Neuroscience* 7, no. 8: 847–854. https://doi.org/10.1038/nn1276.

West, M. J., and A. P. King. 1985. "Social Guidance of Vocal Learning by Female Cowbirds: Validating Its Functional Significance." *Zeitschrift für Tierpsychologie* 70, no. 3: 225–235. https://doi.org/10.1111/j.1439-0310.1985.tb00514.x.

West, M. J., and A. P. King. 1987. "Settling Nature and Nurture Into an Ontogenetic Niche." *Developmental Psychobiology* 20, no. 5: 549–562. https://doi.org/10.1002/dev.420200508.

West, M. J., A. P. King, and M. A. Duff. 1990. "Communicating About Communicating: When Innate Is Not Enough." *Developmental Psychobiology* 23, no. 7: 585–598. https://doi.org/10.1002/dev.420230705.

West-Eberhard, M. J. 2003. *Developmental Plasticity and Evolution*. Oxford University Press. https://doi.org/10.1093/oso/9780195122343.001.0001.

Wilson, J. G. 1959. "Experimental Studies on Congenital Malformations." *Journal of Chronic Diseases* 10, no. 2: 111–130. https://doi.org/10.1016/0021-9681(59)90026-8.

Yanai, I., and M. Lercher. 2020. "Renaissance Minds in 21st Century Science." *Genome Biology* 21, no. 1: 67. https://doi.org/10.1186/s13059-020-01985-6.

Zupic, I., and T. Čater. 2015. "Bibliometric Methods in Management and Organization." *Organizational Research Methods* 18, no. 3: 429–472. https://doi.org/10.1177/1094428114562629.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Table S1:** Keywords (stem words) used to assign publications citing West and King (1987) to different fields, alongside their corresponding weight. The weight was decided based on how important the different words are for identifying fields. Note that a few keywords are not stem words. This was because otherwise they could be confused with keywords from other fields (e.g., "evolutionary psycho-" could be processed as the biology keyword "evolution-"). **Table S2:** Topics of publications citing West and King (1987) within each field, along with their 10 main keywords (from highest to lowest probability) and their interpretation. **Table S3:** Basic properties of the network's CNM fast greedy communities. **Table S4:** Basic properties of the network's Louvain communities. **Figure S1:** Fields to which the publications that have cited West and King (1987) belong. A

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publication that didn't fit any of these categories is excluded ($N = 196$). **Figure S2:** Basic properties of the network. (A) Degree distribution of the network, displaying the frequencies of different degree ($<k>$) values. The departure from the standard heavy-tailed distribution shape of most networks is explained by the fact that the present network was constructed from references with weight ≥ 3, which removes most low-degree nodes. (B) Correlation between degree and strength of nodes in the network. Note that nodes with particularly high degree values have strength values above the least-squares line, indicating that they are particularly intensely connected to other works. **Figure S3:** The network works' weighted betweenness centrality and strength, indicating their connection across communities (betweenness; with communities inferred from the CNM algorithm), and their overall influence (strength; i.e., how frequently and intensely they connect to other works in the network). The quadrants help visualize the most important works (upper right quadrant; see text below for a context). **Figure S4:** Network structure using the Louvain community detection algorithm. Since the numbering of the communities is arbitrary, the color granted to each of them is the one that roughly corresponds most to those resulting from the CNM algorithm, which in turn comes from the palette used for the field-level analyses. **Figure S5:** Authors' average semantic interdisciplinarity (mean normalized Shannon index over frequencies of psychology, biology, philosophy, and medicine keywords in their publications' titles and abstracts), and weighted intensity of West and King (1987) citation (fractional contributions citing this work). Only authors with one or more weighted contributions are displayed. The authors with more bridge-building potential (Griffiths, Stotz, and Lickliter) are signaled.

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